

JONATHAN WITCOSKI

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Washington, DC metropolitan area (previously Reston, Virginia) | Citizenship: United States

SUMMARY OF QUALIFICATIONS

Geographic Information Systems (GIS) developer with more than 15 years of experience designing, building, and supporting enterprise geospatial applications for federal agencies and regulated industries. Skilled in Esri ArcGIS Enterprise, ArcGIS Online, ArcGIS Pro, Python, R, JavaScript, Structured Query Language (SQL), SQL Server, PostgreSQL/PostGIS, REST application programming interfaces (APIs), and automated extract-transform-load (ETL) workflows. Experience includes public health surveillance mapping, utility GIS migration, census geography production, and emergency management support for the Department of Homeland Security (DHS) and the Federal Emergency Management Agency (FEMA).

WORK EXPERIENCE

Geographer

Employer: DRT Strategies (Centers for Disease Control and Prevention) | Location: Remote

Dates: 03/2022 – Present | Hours per week: 40 | Supervisor: Available upon request | May contact: Yes

- Develop web mapping applications and dashboards using ArcGIS Enterprise, JavaScript, React, and Power BI for Centers for Disease Control and Prevention (CDC) public health surveillance programs.
- Use R and Python to build PMTiles and GeoParquet datasets that feed dashboards and web maps.
- Build SQL Server and PostgreSQL/PostGIS databases and Python/SQL ETL pipelines for recurring geospatial updates.

Founder

Employer: Vector Scope AI LLC | Location: Remote

Dates: 10/2025 – Present | Hours per week: 15 | Supervisor: Self | May contact: Yes

- Build ArcGIS-connected web tools and Python pipelines for map editing, validation, and data extraction.
- Deliver Global Ski Atlas (3,000+ ski areas) as a public web mapping application with automated ETL.

GIS Data Engineer and Scrum Master

Employer: Saicon (National Grid) | Location: Remote

Dates: 05/2021 – 03/2022 | Hours per week: 40 | Supervisor: Available upon request | May contact: Yes

- Migrated large utility GIS datasets to cloud-hosted ArcGIS Enterprise using Python and SQL.
- Led Agile delivery of migration milestones and spatial data quality checks.

Geographer

Employer: U.S. Department of Commerce, U.S. Census Bureau | Location: Suitland, Maryland

Dates: 11/2016 – 03/2021 | Hours per week: 40 | Series/Grade: Available upon request

Supervisor: Available upon request | May contact: Yes

- Developed Python tools and ArcGIS API workflows for census geography production.
- Automated spatial processing workflows across large national datasets.

GIS Systems Administrator and Software Engineer

Employer: C2 Solutions Group Inc. | Location: Reston, Virginia

Dates: 05/2014 – 11/2016 | Hours per week: 40 | Supervisor: Available upon request | May contact: Yes

- Administered ArcGIS Server and Portal deployments backed by SQL Server for federal clients.
- Built secure web mapping applications used by hundreds of concurrent users.

Geospatial Analyst

Employer: Booz Allen Hamilton (DHS/FEMA) | Location: Philadelphia, Pennsylvania and Arlington, Virginia

Dates: 10/2009 – 05/2014 | Hours per week: 40 | Supervisor: Available upon request | May contact: Yes

- Built maps, spatial analysis workflows, and web GIS tools for FEMA and DHS disaster response operations.
- Supported situational awareness and decision-making during major disaster events.

Earlier GIS Roles (2007 – 2009)

Employer: Tyco Telecommunications; WDG Location Consulting; Philmont Scout Ranch

Dates: 05/2007 – 05/2009 | Hours per week: 40 (full-time and seasonal roles)

Supervisor: Available upon request | May contact: Yes

- GIS solutions development, location consulting analysis, and field GIS support.

EDUCATION

Master of Science, Geography | University of Tennessee, Knoxville | 05/2007

Bachelor of Arts, Geography and Anthropology; Minor in Geographic Information Systems | Pennsylvania State University | 05/2005

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services | 03/2021

TECHNICAL SKILLS

ArcGIS Enterprise, ArcGIS Online, ArcGIS Pro, ArcGIS API for JavaScript, Spatial Analyst, Python, R, JavaScript, SQL, SQL Server, PostgreSQL/PostGIS, GeoParquet, PMTiles, REST APIs, web services, AWS, Azure, Docker, Jenkins, GitLab, continuous integration/continuous delivery (CI/CD), React, Power BI, Agile/Scrum